

TIVAT, Montenegro

WMO: 134570

Lat: 42.405N

Lon: 18.723E

Elev: 6

StdP: 101.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-1.0	-13.7	1.1	4.0	-11.1	1.4	5.1	11.1	7.3	9.6	7.9	0.9	80	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	34.5	22.4	33.2	22.4	32.0	22.1	24.5	31.3	23.8	30.8	23.0	30.1	2.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	16.9	28.7	21.3	15.9	27.8	20.6	15.3	27.1	74.1	31.4	71.0	31.1	68.1	30.1	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	6.8	5.6	-4.8	36.7	1.3	1.3	-5.7	37.6	-6.5	38.4	-7.3	39.1	-8.3	40.0		
(5)			-5.7	26.2	1.3	1.0	-6.7	26.9	-7.4	27.4	-8.2	28.0	-9.2	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	7.4	8.2	10.4	13.6	18.0	22.1	24.7	24.7	20.4	16.4	12.3	8.7		
	DBStd	6.71	2.89	2.91	2.69	2.75	2.64	2.89	2.12	2.21	2.56	2.56	3.17	3.02		
	HDD10.0	262	90	65	28	3	0	0	0	0	0	0	13	62		
	HDD18.3	1617	339	285	247	142	40	4	0	0	9	69	182	300		
	CDD10.0	2312	8	13	39	113	247	364	456	456	313	198	83	21		
	CDD18.3	625	0	0	0	2	28	118	198	198	72	9	1	0		
	CDH23.3	7293	0	1	1	28	314	1402	2459	2364	656	67	1	0		
	CDH26.7	3036	0	0	0	3	51	527	1153	1134	165	2	0	0		
(14) Wind	WSAvg	1.7	2.0	2.2	2.2	1.7	1.5	1.4	1.5	1.4	1.5	1.4	1.7	2.0		
(15) Precipitation	PrecAvg	2091	235	210	196	194	111	76	49	74	161	225	288	273		
	PrecMax	3031	528	422	471	350	211	247	238	253	397	503	545	422		
	PrecMin	1189	10	0	3	60	16	9	7	14	22	9	80	67		
	PrecStd	446	134	100	95	77	53	56	58	66	95	109	97	114		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	18.8	21.1	26.4	30.0	34.8	36.1	36.2	32.0	26.6	22.2	17.9		
		MCWB	11.9	12.0	12.7	16.6	20.0	22.9	22.8	22.3	20.9	19.2	16.2	13.2		
	2%	DB	15.0	16.8	19.1	23.6	28.1	32.9	34.2	34.9	30.1	25.1	20.7	16.6		
		MCWB	11.7	11.4	12.1	15.1	19.1	22.7	22.6	22.2	21.0	18.5	16.3	13.2		
	5%	DB	14.0	15.2	17.9	21.8	26.6	31.2	33.1	33.3	28.6	23.9	19.6	15.4		
		MCWB	11.0	10.7	12.1	14.3	18.6	22.0	22.5	22.3	20.4	17.8	15.6	12.3		
	10%	DB	12.8	14.0	16.4	20.1	25.1	29.9	32.0	32.0	27.1	22.3	18.1	14.2		
		MCWB	9.6	10.2	11.3	13.8	18.1	21.4	22.1	22.2	19.9	17.2	14.5	11.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.8	14.8	17.5	21.5	25.0	25.5	25.6	23.7	21.0	18.5	15.6		
		MCDB	15.2	15.9	17.9	23.6	27.1	31.2	32.4	31.6	28.7	24.0	20.5	16.8		
	2%	WB	12.7	12.7	13.9	16.4	20.2	23.8	24.5	24.5	22.4	20.0	17.4	14.2		
		MCDB	14.3	15.1	17.1	21.8	26.2	30.8	31.6	31.3	27.5	23.1	19.7	15.8		
	5%	WB	11.6	11.9	13.1	15.4	19.4	22.9	23.7	23.7	21.5	19.0	16.1	13.2		
		MCDB	13.3	14.4	16.3	20.0	25.3	29.7	31.0	30.9	26.7	22.4	18.6	14.8		
	10%	WB	10.6	11.0	12.3	14.6	18.5	22.0	22.9	22.9	20.7	18.1	15.1	12.2		
		MCDB	12.1	13.5	15.5	18.7	23.9	28.7	30.3	30.2	25.7	21.5	17.5	13.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.0	10.4	11.0	11.8	12.5	13.2	13.4	12.0	11.4	10.1	9.4		
		MCDBR	10.8	12.0	13.1	13.7	14.2	14.4	15.1	15.4	13.8	13.3	11.4	10.5		
		MCWBR	7.6	7.8	7.7	7.0	7.0	6.5	6.5	6.7	6.9	7.6	7.2	7.3		
		MCDBR	8.3	9.8	11.0	12.2	13.3	13.5	13.9	14.0	12.3	10.8	9.2	7.2		
	5% WB	MCWBR	6.6	7.1	7.4	7.0	7.2	6.8	7.1	7.3	7.1	7.0	6.4	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.340	0.365	0.410	0.463	0.450	0.435	0.434	0.444	0.423	0.396	0.367	0.340		
	taud		2.426	2.339	2.227	2.103	2.205	2.298	2.295	2.261	2.311	2.364	2.393	2.430		
	Ebn at Noon		793	827	827	808	830	842	838	816	806	782	749	756		
	Edh at Noon		79	100	125	153	141	129	128	129	114	97	81	72		
(44) All-Sky Solar Radiation	RadAvg		1.65	2.38	3.65	4.85	6.03	6.93	7.14	6.32	4.53	3.09	1.91	1.50		
	RadStd		0.19	0.38	0.51	0.51	0.47	0.43	0.38	0.39	0.33	0.30	0.19	0.27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-1.93	N/A	N/A	N/A	N/A	-80	+100	N/A
(47) Regional (22 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-106	+100	N/A

Nomenclature: See separate page